

ORCA

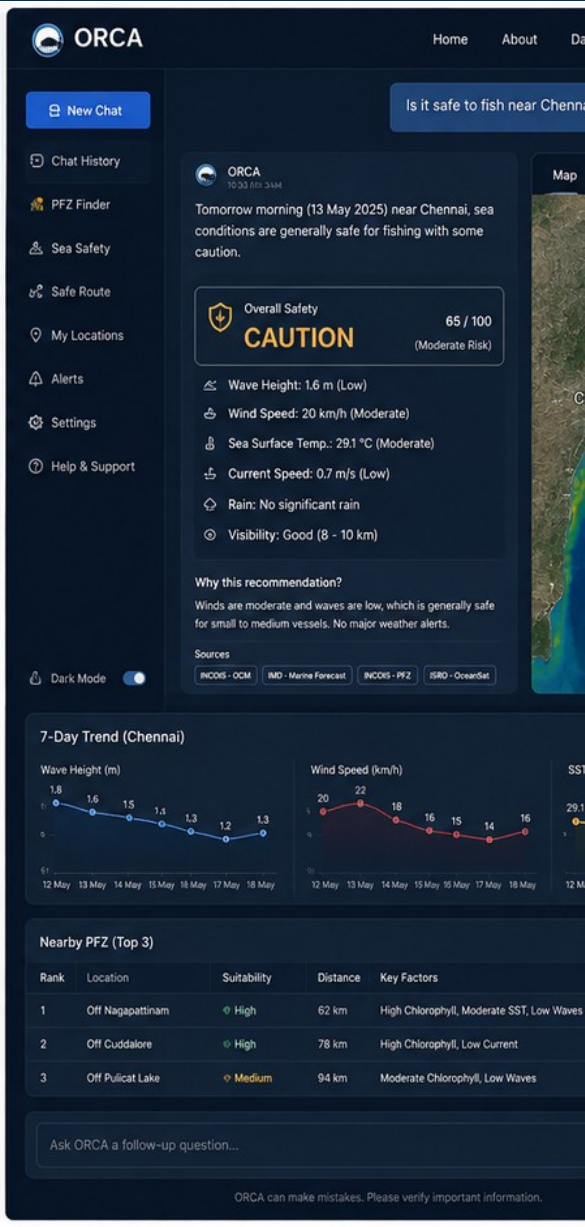
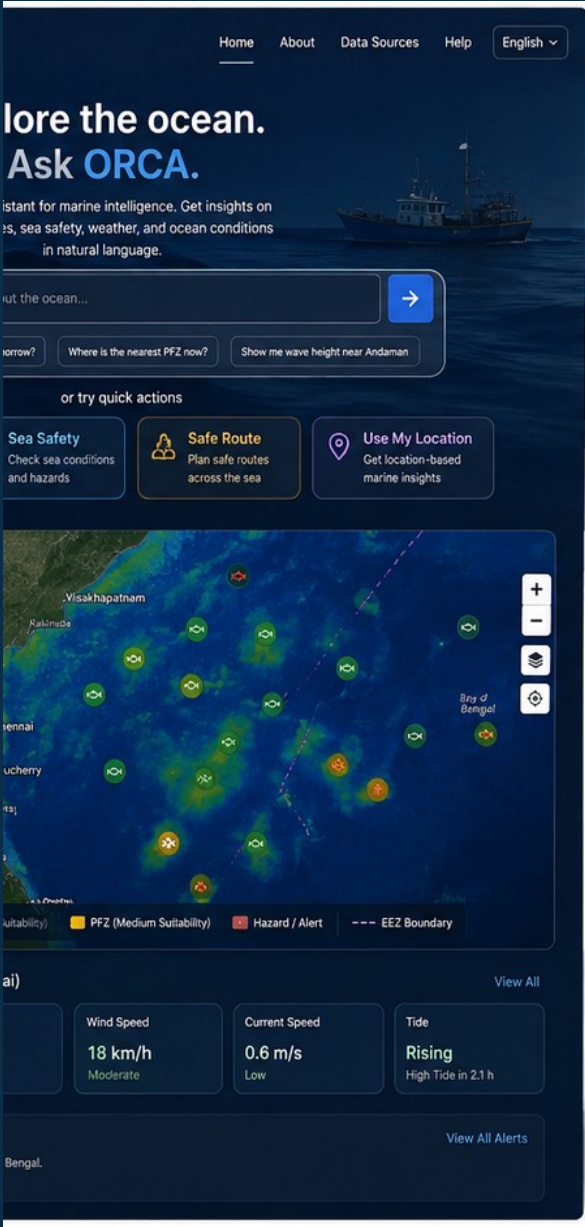
Agentic AI for Ocean Intelligence

Ask about the ocean in simple language. ORCA connects the right information, analysis and maps to help you decide what to do next.

SIH26176 • SOFTWARE • MISCELLANEOUS

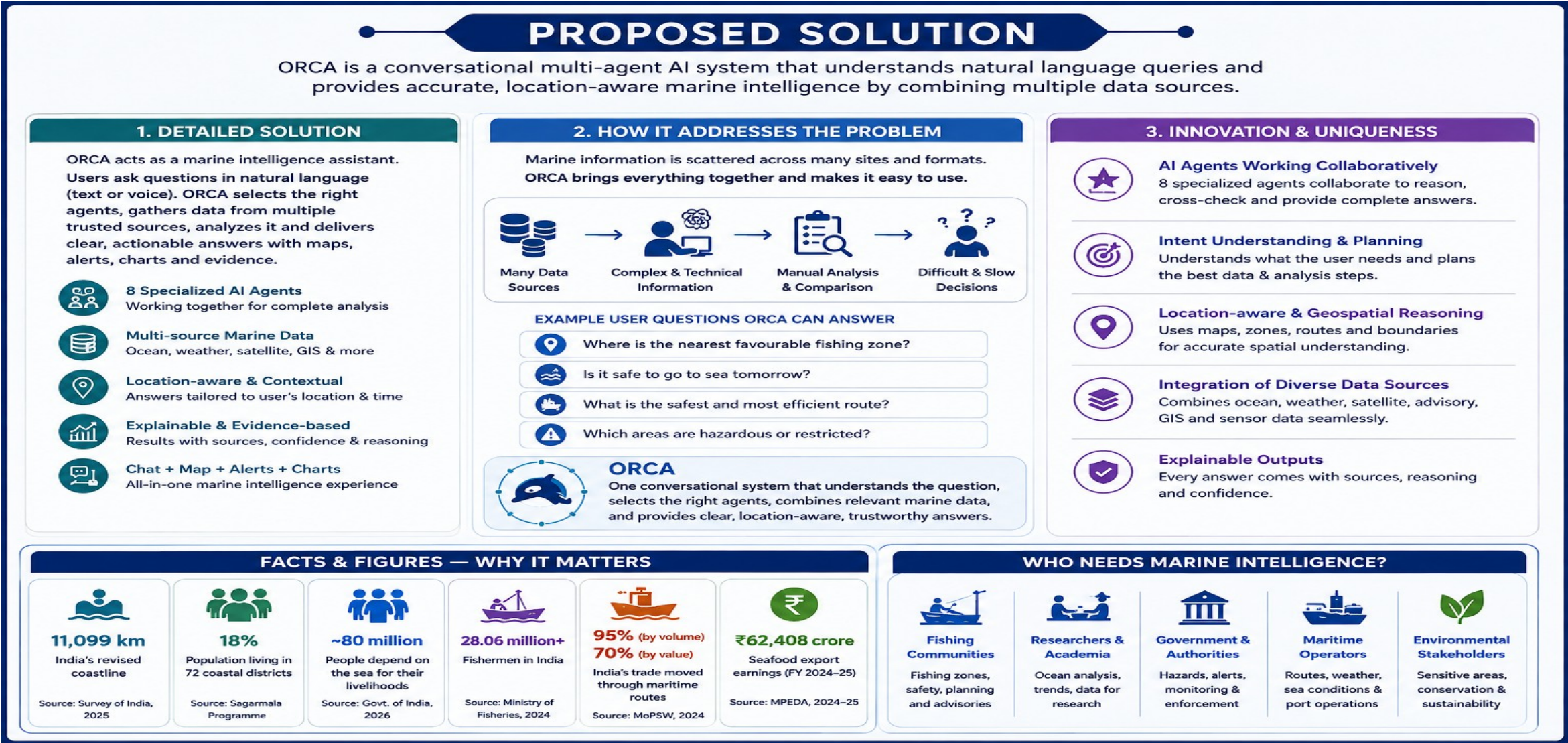
Problem: ORCA Marine EcOsysteem Reasoning with Collaborative Agents

Team ID: _____ Team Name: CodeCatalysts



Proposed Solution

Exact generated solution visual — kept as the main design reference.

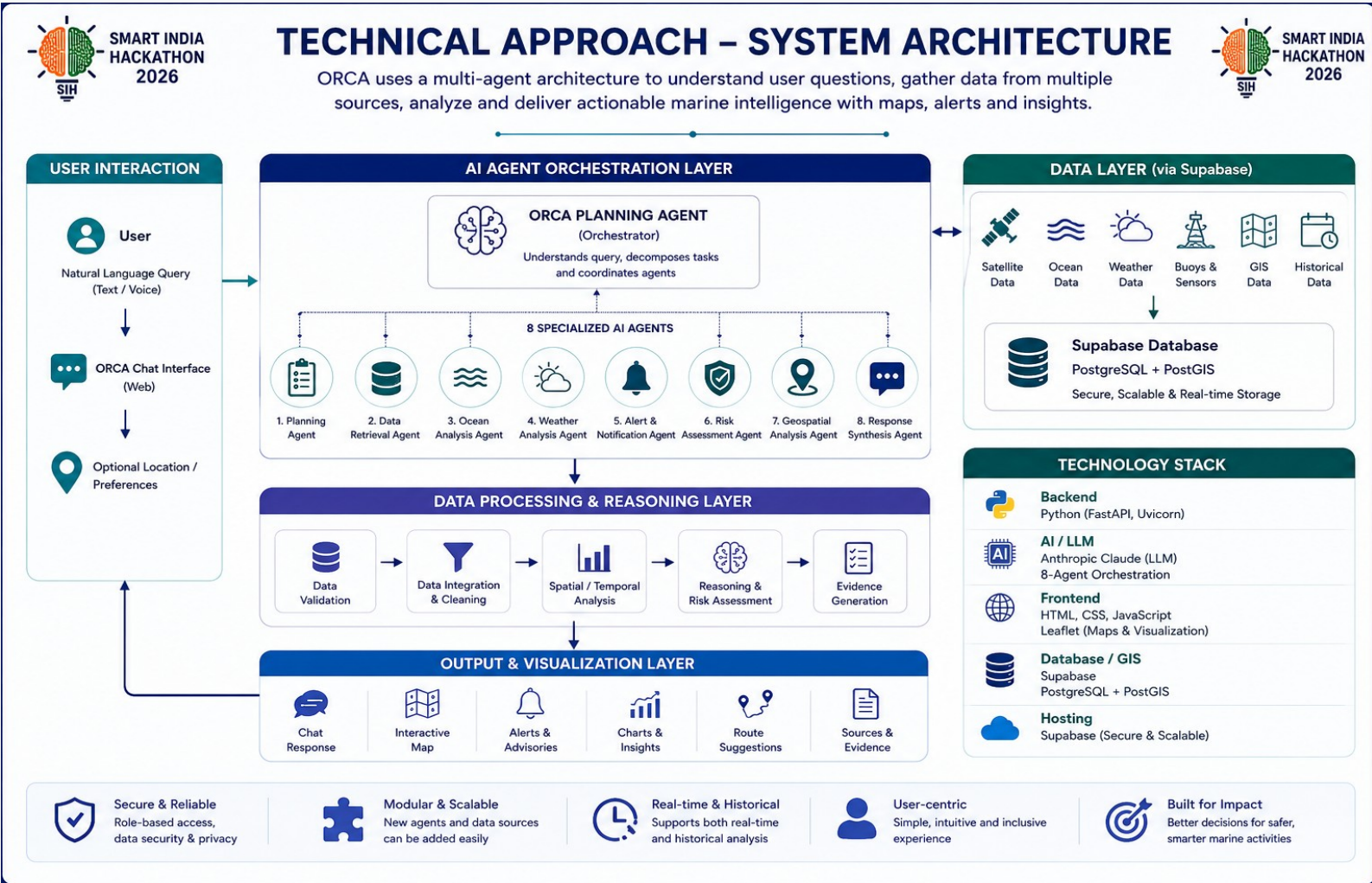


Technical Approach

Editable implementation flow on the left; exact generated architecture visual on the right.

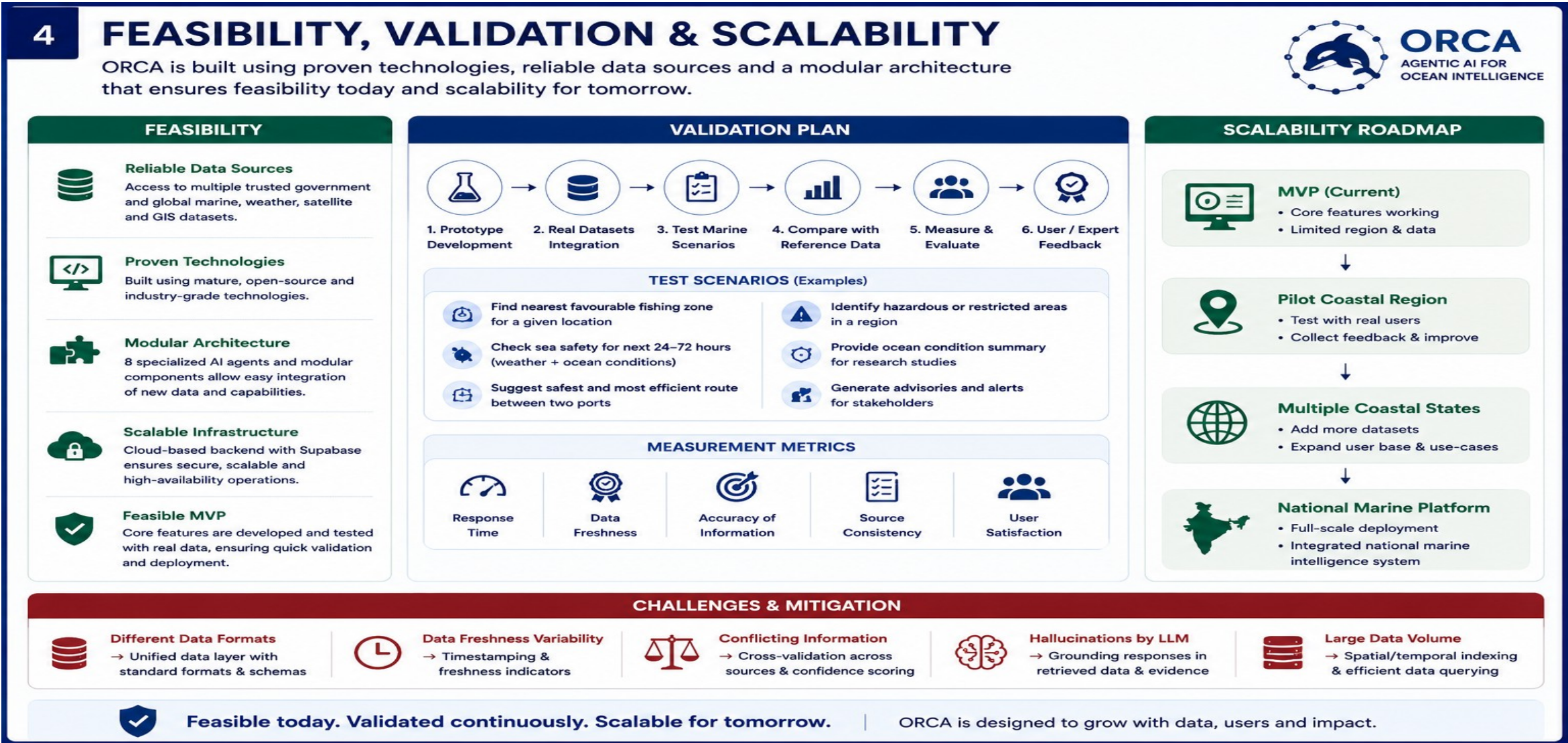
IMPLEMENTATION FLOW

- 1 USER INPUT**
Question + optional location
- 2 PLANNING AGENT**
LLM understands the request + builds the task plan
- 3 8 SPECIALISED AGENTS**
Work together on the required task
- 4 DATA**
Marine + weather + satellite + GIS
- 5 VALIDATE & ANALYSE**
Check source, time, location + relationships
- 6 REASON & ASSESS**
Combine results, constraints and risk
- 7 OUTPUT**
Chat + map + alerts + route + evidence



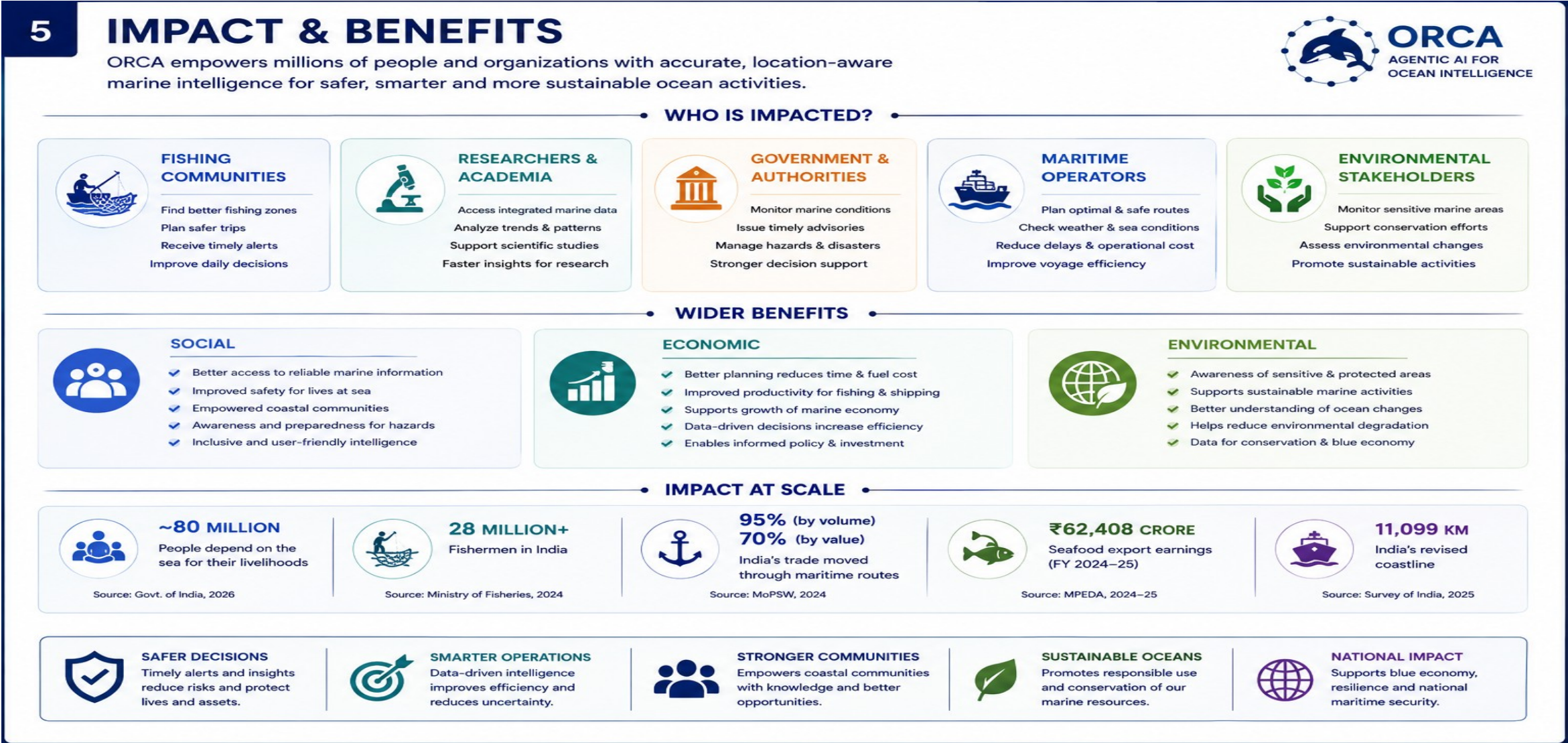
Feasibility, Validation & Scalability

Exact generated feasibility visual — this is our working slide design.



Impact & Benefits

Exact generated impact visual — with stakeholder groups, wider benefits and scale.



Research & References

Exact generated research/reference visual — source names can be finalized later.

6

RESEARCH AREAS, DATA SOURCES & REFERENCES

ORCA is built on proven research, trusted marine data sources and open technologies.



ORCA
AGENTIC AI FOR
OCEAN INTELLIGENCE

RESEARCH AREAS



Marine Earth Observation

Satellite remote sensing for sea surface parameters, chlorophyll, SST, waves, currents and ocean color.



Ocean & Weather Forecasting

Numerical models and forecasts for weather, waves, winds, temperature and extreme events.



In-situ Observations

Buoys, tide gauges and sensors providing real-time marine and meteorological data.



Geospatial Analysis

Spatial data processing, GIS, geoprocessing and location-aware decision support.



Multi-Agent AI & LLMs

Agentic AI orchestration, reasoning, planning and natural language understanding.



Decision Support Systems

Risk assessment, alerts, recommendations and visualization for better decision-making.

DATA SOURCE CATEGORIES



Satellite & Ocean Observations

Sea surface temperature, ocean color, winds, sea level, currents, wave height, SST, chlorophyll, salinity, etc.



Weather & Ocean Forecasts

Weather forecasts, wave forecasts, wind forecasts, cyclones, swell and storm surge predictions.



Buoys & In-situ Sensors

Real-time data from buoys, tide gauges, ADCPs, wave riders, met stations, etc.



Marine Advisories & Alerts

Fisheries advisories, navigational warnings, cyclone alerts, coastal warnings, etc.



GIS & Geographic Boundaries

Coastline, EEZ, maritime boundaries, ports, fishing zones, marine protected areas, bathymetry, etc.



Historical Datasets

Historical satellite archives, reanalysis data, past storms, long-term climate and ocean records.

KEY REFERENCES & RESOURCES



INCOIS – Indian National Centre for Ocean Information Services

<https://www.incois.gov.in>

Ocean data, forecasts, advisories, buoys & model outputs



NOAA / CoastWatch

<https://coastwatch.noaa.gov>

Satellite ocean data and imagery



NOAA OceanWatch

<https://oceanwatch.noaa.gov>

Real-time ocean conditions and visualizations



WaveWatch III

<https://polar.ncep.noaa.gov/waves>

Global wave model data



Leaflet – JavaScript Library

<https://leafletjs.com>

Open-source library for interactive maps



Supabase

<https://supabase.com>

Backend, authentication, real-time database and storage



PostGIS (Spatial for PostgreSQL)

<https://postgis.net>

Spatial database for GIS operations



Our Approach

ORCA combines cutting-edge research, reliable multi-source data and **agentic AI** to deliver accurate, explainable and location-aware marine intelligence for everyone.



Reliable Data

Trusted sources and real-time updates



Evidence Based

Every output is backed by data and sources



Open & Scalable

Built on open tools and modular architecture



Research Driven

Continuously improving with latest research

“ Bringing together data, intelligence and technology for a safer, smarter and sustainable ocean future. ”